Literature-Based Analytical Reference. To establish an independent reference solution, we reproduced the two-dimensional transient heat-conduction study of Hsu, Tu, and Chang [1], which considers a rectangular domain with space- and time-dependent Dirichlet boundary conditions. Their method decomposes the temperature field into two one-dimensional subsystems by superposition, applies shifting functions to convert the nonhomogeneous boundary conditions into homogeneous ones, and then obtains the temperature through truncated sine-eigenfunction expansions. We implemented their analytical expressions for the parabolic-boundary example and evaluated the center temperature using the same truncation levels, $m = n = \{1, 3, 5, 10, 20\}$. The calculated values for Table 1 closely reproduce the reported temperatures and convergence behavior, confirming that approximately 5-10 terms are sufficient for the published precision. The values printed in Tables 2 and 3 were also preserved; however, direct evaluation of the published equations with the stated decay constants does not reproduce those two tables, indicating an inconsistency in the source article. The original study reports eight time instances between tau = 0 and 1.2; we extended each boundary-condition case to 40 uniformly spaced instances over the same interval to provide denser analytical reference data. In the attached project archive, the implementation is located at `Literature_results/mdpi_416_series.py`, while the reproduced and extended tables are provided under `Literature_results/mdpi_416_tables/`.

References

[1] H.-P. Hsu, T.-W. Tu, and J.-R. Chang, "An Analytic Solution for 2D Heat Conduction Problems with General Dirichlet Boundary Conditions," *Axioms*, Vol. 12, No. 5, Article 416, 2023. doi:10.3390/axioms12050416.